Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Lee, Te-Won Ph.D.		
Project Title:	Automatic Speech Recognition in Noisy Environments - Human Factors Approach		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tlee@softmax.com	**Fax:**	FY 619-839-3119
PI Organization Type:	INDUSTRY	**Phone:**	858-736-5300
Organization Name:	SoftMax, Inc		
PI Address 1:	6885 Flanders Drive, Suite A		
PI Address 2:			
PI Web Page:	www.softmax.com		
City:	San Diego	**State:**	CA
Zip Code:	92121	**Congressional District:**	49
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	03/10/2003	**End Date:**	04/10/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1479		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of the first year was the optimization of ICA technology by SoftMax, Inc. for speech enhancement in noisy shuttle (ISS) environment. To achieve the task goals, we have worked on a specific task related to voice recgontiion for the robonaut project. A commercial recognition system is enhanced by our ICA technology as a front filter. We have also optimized the vocabulary to minimize errors. In ISS voice recognition capability enhances the human performance. Accurate and reliable recognition is necessary to deploy the system. Our technology yields significant improvements over the existing technology.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The focus of the application has been on Robonaut Command and Control via voice recognition. Evaluation plans are considered for on ground applicability and designs.
Task Progress:	Activities: - Meetings with JSC team in June 2003 - Robonaut vocabulary specification - Meetings with JSC team in December 2003 - Obtained sound samples from space station - Tests and initial evaluation in acoustic rooms at JSC Results: - Initial optimization of Robonaut vocabulary with linguist consultant - Developed preliminary system with Via Voice interface - Obtained initial high accuracy in evaluation at acoustic chamber at JSC - Via Voice is not recommended for task because it is too sensitive to noise
Bibliography Type:	Description: (Last Updated: 06/13/2005)
Abstracts for Journals and Proceedings	Lee, Te-Won "Automatic Speech Recognition in Noisy Environments – A Human Factors Approach" Automatic speech recognition system will facilitate the effectiveness of the human system interface by providing hands-free command and control functionality for time and safety critical operations such as robotics operations during ISS assembly, shuttle remote manipulator system/closed circuit television system operations, and protein crystal growth experiments or dissection tasks using the glove boxes. The goal of this proposal is to develop a new state-of-the-art automatic voice recognition system that works accurately in noisy space flight environments. The targeted system will achieve high recognition accuracy and therefore allow robust identification and processing of voice commands. The new technology to be implemented in this project is Independent Component Analysis (ICA) that has been recently used in hands-free voice command systems in noisy environments. ASR, ICA , Jan-2004